Cyber Physical Robotics

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Cyber Physical Robotics

Today:

- Markov-Chain Monte-Carlo (MCMC) applied to the simulation of CPRs
  - Specifically, combined with Petri Nets
- System design using Petri Nets

Stochastic Petri Nets

Stochastic Petri Nets

- MCMC is one type of stochastic techniques
  - Remember: this is a design tool

  The goal is to evaluate how (and if) the system will work before spending time and money building it

Stochastic Petri Nets

Why?

- Ideally, we'd simulate all possible cases (in practice, this is impossible)
- Or use algebraic techniques (i.e., calculate probabilities analytically)

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Why?

- All the probability mathematics you've learned only work under IID assumption
- (events are Independent and Identically Distributed random variables)
- If we're looking at the evolution of a concurrent system:
  - The probability of event occurrence at time t
  - Depends on system evolution during time $0, t-1$
- Events are not IID!

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Thus, MCMC

- A way to obtain statistical assurances about dependent events
- E.g., what is the chance my system will evolve towards a certain state
- Case study:
  - Real system design (simplified): furniture factory

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MCMC recap

- Monte-Carlo Method:
  - Random-numbers based simulation to obtain knowledge
  - E.g., value of Pi?
  - Let's see code

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MCMC recap

- Markov-Chain Monte-Carlo method:
  - What is *possible* to simulate via Monte-Carlo (the **next state**) depends on the current state

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MCMC recap

- Example: ant in search of sugar
  - It will *probabilistically* head in the same direction of sugar
  - But there's a chance it will turn in the wrong direction
  - Let's see the code

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Furniture factory

- Wood comes in, shelves go out
- Production (mostly) automated by robots
- Some manual work

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Furniture factory: 3 production steps
1

- Tree trunks delivered from storage by truck (time varies)
- Once in, cut into long boards (robot)
- Long boards placed in buffer area (robot)

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Furniture factory: 3 production steps
2

- Long boards from buffer cut into two smaller parts (human)
- Only if mechanical saw available (may be used by other people, variable)
- Smaller boards divided into two buffer areas

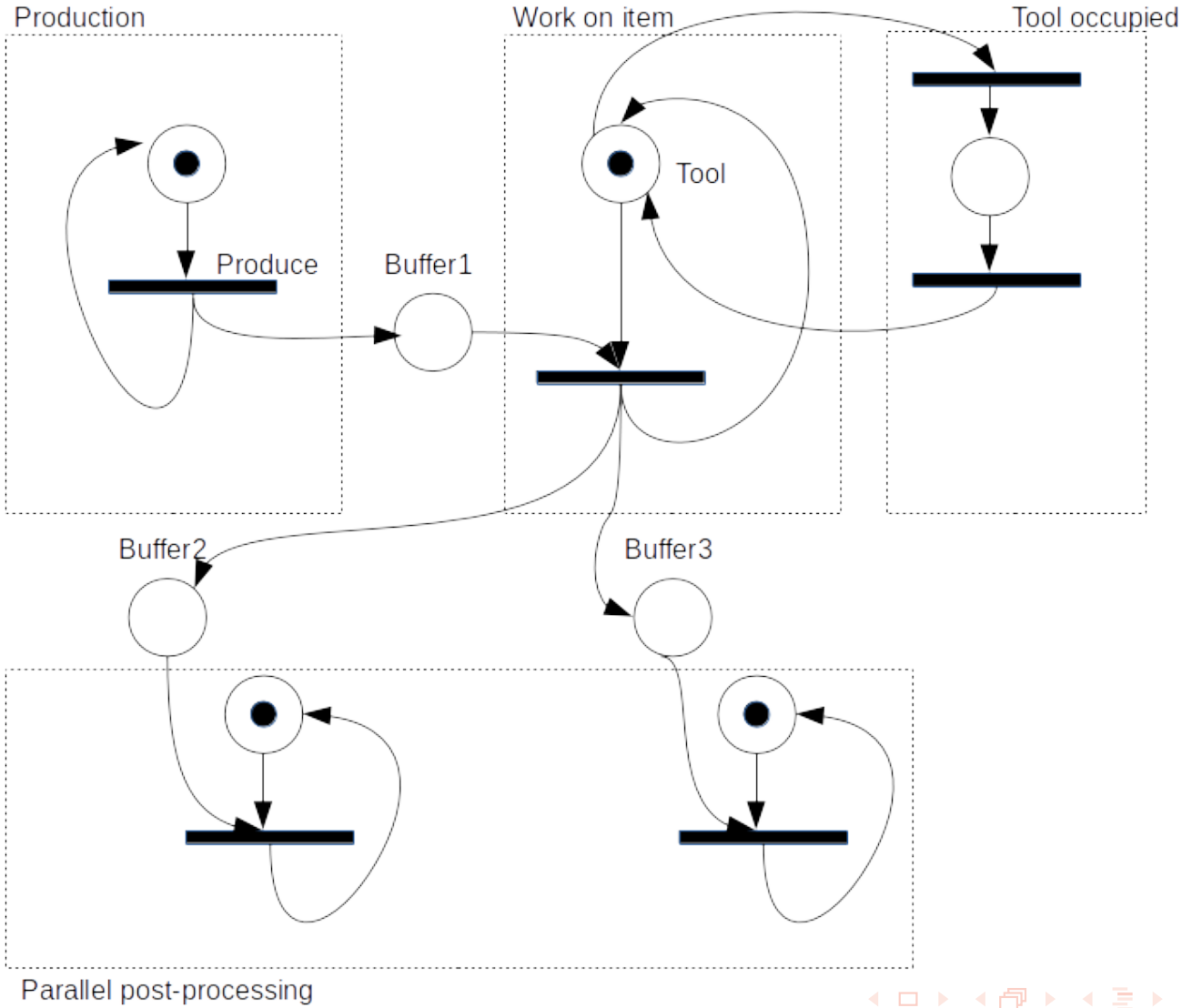
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Furniture factory: 3 production steps
3

- Small boards assembled into shelves (robots)
- 2 parallel production lines

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Furniture factory: 3 production steps



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Furniture factory: **key design question**

- How big do the buffer areas need to be?

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Furniture factory: **key design question**

- How big do the buffer areas need to be?
- Equivalent to:
  - On average, at any point in time, how many long/small boards are in the buffer areas?

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Furniture factory: **key design question depends on**

- How often tree trunks are delivered

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Furniture factory: **key design question depends on**

- How often tree trunks are delivered
- How fast they are cut into long boards

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Furniture factory: **key design question depends on**

- How often tree trunks are delivered
- How fast they are cut into long boards
- How often the mechanical saw is unavailable

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Furniture factory: **key design question depends on**

- How often tree trunks are delivered
- How fast they are cut into long boards
- How often the mechanical saw is unavailable
- If available, how fast they are processed into small boards

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Furniture factory: **key design question depends on**

- How often tree trunks are delivered
- How fast they are cut into long boards
- How often the mechanical saw is unavailable
- If available, how fast they are processed into small boards
- How fast they are assembled into shelves

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Furniture factory: **key design question**

- How big do the buffer areas need to be?

How do we figure this out?

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Furniture factory: **key design question**

- How big do the buffer areas need to be?

How do we figure this out?

MCMC on Petri Nets

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Why MCMC on Petri Nets?

- Process is concurrent: need to model it using a concurrent formalism
- Process has inter-dependent events

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Let's put this into practice

- Implement a Petri Net that models the system
- Implement a MCMC simulation for it